



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

15-08-2021

JEE(Main + Advanced) : Enthusiast Course (PHASE : TEAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	B	A	A	B	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,D	A,B,C,D	A,C	A,C,D	A,B	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	5.26	1.38 to 1.39	2.12 to 2.13	36.00	1000.00	14.08

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	D	C	A	D	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C,D	A,B	B,C	A,D	A,B	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	7.00	3.00	68.14 to 68.15	2.00	3.00	3.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	B	C	D	A	A
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B	A,C,D	B,C	A,B,C,D	A,B,C,D	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	15.00	1.00	3.14	1.50	0.50	9.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (B)**

$$I \propto \frac{1}{r^2}$$

$$SL = 20 \log_{10} = \frac{I}{I_0}$$

2. **Ans (B)**

$$mL + m \times s (100 - 80)$$

$$= 1.12 \times 5 \times (80 - 15)$$

3. **Ans (A)**

$$K_{\max} = \frac{hc}{\lambda} - \frac{hc}{\lambda_{th}} = eV_s = eEd$$

$$\Rightarrow \lambda = 413.3 \text{ nm}$$

4. **Ans (A)**

$$n_{C_2} = 10 \Rightarrow n = 5$$

$$\frac{E_0}{25} = E_0 2^2 \left(\frac{1}{m^2} - \frac{1}{5^2} \right)$$

$$m = 2 \text{ or } 3$$

$$\text{for } m = 2, \frac{144}{25} = z^2 \left(\frac{1}{4} - \frac{1}{25} \right) = \frac{21}{100} z^2$$

$$\text{for } m = 3$$

$$\frac{144}{25} = z^2 \left(\frac{1}{9} - \frac{1}{25} \right) \quad z \text{ is not integer}$$

$$z^2 = 81 \Rightarrow z = 9$$

5. Ans (B)

$$y \text{ cm} = (\mu_1 - \mu_2) \frac{tD}{d} = \frac{5\lambda D}{d}$$

$$t = \frac{5\lambda}{(\mu_1 - \mu_2)} = \frac{0.48 \times 5}{0.3} = 8\mu\text{m}$$

6. Ans (C)

$$\lambda = \frac{h}{p_m}$$

$$1.005\lambda = \frac{h}{p_0 - p_m}$$

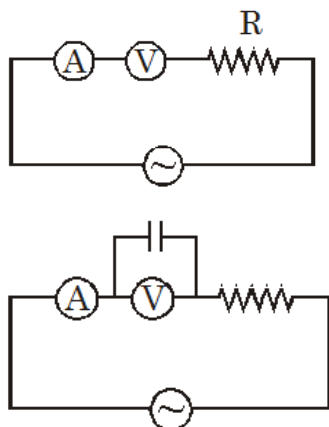
$$\Rightarrow 1.005 = \frac{p_0}{p_0 - p_m}$$

$$\Rightarrow 0.005p_0 = 1.005 p_m$$

$$p_0 = 201 p_m$$

SECTION-I (ii)

7. Ans (A,D)

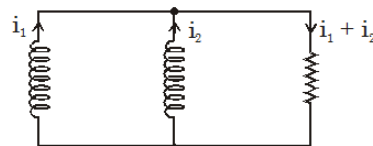


$$i = \frac{V_0}{R_A + R_V + R}$$

$$i' = \frac{V_0}{R_A + R + \frac{R_V \times j \times C}{R_V - j \times C}}$$

$$i' < i \Rightarrow V'_0 > V_0$$

8. Ans (A,B,C,D)



at any instant

$$x = -1 \frac{di_1}{dt} = -2 \frac{di_2}{dt} = (i_1 + i_2) \times 100$$

$$\Rightarrow \left. \begin{aligned} -\frac{x}{2} &= \frac{di_2}{dt} \\ -x &= \frac{di_1}{dt} \end{aligned} \right\} = -\frac{3x}{2} = \frac{d}{dt}(i_1 + i_2)$$

$$= -150(i_1 + i_2)$$

$$\Rightarrow \frac{i_1 + i_2}{6} = e^{-150t}$$

$$\text{at } t = \infty, i_1 + i_2 = 0$$

$$q_R = 6 \int_0^\infty e^{-150t} dt$$

$$= \frac{6x}{-150} e^{-150t} \Big|_0^\infty$$

$$= \frac{1}{25} = 0.04C$$

$$\text{In steady state } i_1 + i_2 = 0 \Rightarrow i_1 = -i_2$$

$$\frac{di_1}{dt} = \frac{2di_2}{dt}$$

$$\Rightarrow i_1 - 3 = 2(i_2 - 3)$$

$$i_1 - 2i_2 = -3$$

$$3i_1 = -3 \Rightarrow i_1 = -1A, i_2 = 1A$$

9. Ans (A,C)

Part A & B

$$\ell = 20 \text{ mm} \Rightarrow x \text{ in rubber band}$$

$$= 40 \text{ mm} - 39 \text{ mm} = 1 \text{ mm}$$

$$\Rightarrow F = 0.1 \text{ N} \times 2$$

$$2 \times 0.1 = kx = 200x$$

$$x = \frac{0.1}{100} \times 1000 \text{ mm} = 1 \text{ mm}$$

$$\text{Natural length} = 21 \text{ mm}$$

Path C & D

$$200 \times x_1 = 100 x_2 \times 4$$

$$x_1 = 2\ell - \ell_{\max}$$

$$x_2 = 4\ell = 39 \text{ mm}$$

$$\Rightarrow x_1 = 2x_2$$

$$2\ell - \ell = 8\ell - 78$$

$$99 = 9\ell \Rightarrow \ell = 11 \text{ mm}$$

10. Ans (A,C,D)

$$y = 2 \sin\left(3\pi t - \frac{3\pi x}{v}\right)$$

$$\Rightarrow \frac{\pi}{8} = \frac{3\pi}{v} + 2\pi x$$

$$\frac{1}{8} = \frac{3}{v} + 2x$$

$$\frac{1}{8} = \frac{2}{\lambda} + 2x$$

11. Ans (A,B)

$$e \times \pi R^2 5 = 4\pi R^2 e \sigma (T^4 - T_0^4)$$

$$T^4 = 300^4 + \frac{1536 \times 10^8}{24} = 145 \times 10^8$$

$$T \approx \sqrt[4]{12} \times 10^2 \approx 346K$$

12. Ans (A,C,D)

$$I = 4I_0 \frac{\cos^2 \pi y d}{\lambda B}$$

$$I = 2 \frac{\cos^2 \pi y}{\beta}$$

SECTION-II

1. Ans (5.26)

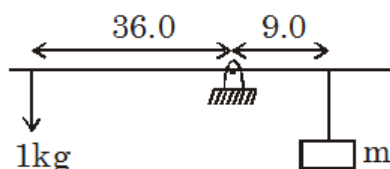
$$mg - B i \ell = ma$$

$$q = C B \ell v$$

$$i = C B \ell a$$

$$a = \frac{mg}{m + C^2 B^2 \ell^2}$$

2. Ans (1.38 to 1.39)



$$m \ell_1 = M \ell_2$$

$$m = \frac{M \ell_2}{\ell_1}$$

$$\frac{\Delta m}{m} = \frac{\Delta \ell_1}{\ell_1} + \frac{\Delta \ell_2}{\ell_2} = \frac{0.1}{9.0} + \frac{0.1}{36.0}$$

3. Ans (2.12 to 2.13)

$$L = 8.5 \frac{\ell}{2} \Rightarrow \lambda = \frac{\sqrt{\frac{y}{p}}}{10^4} = \frac{\sqrt{\frac{2 \times 10^{11}}{8 \times 10^3}}}{10^4}$$

$$= \frac{1}{2} m$$

$$L = \frac{8.5}{4} = 2.12 \text{ to } 2.13$$

4. Ans (36.00)

$$\Delta \ell = \ell \times \Delta \theta = 1.17 \times 10^{-4} \times 100$$

$$= 1.17 \times 10^{-2} \text{ mm}$$

1 msd at 100° C will be

$$1(1 + 2 \times 10^{-5} \times 100)$$

$$= 1.002 \text{ mm}$$

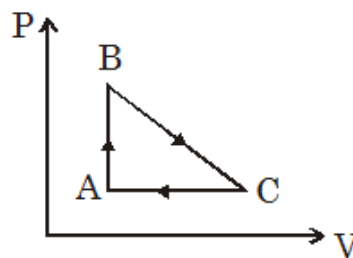
$$\Delta \ell' = 1.17 + 1.17 \times 10^{-2} - 1.002$$

$$= 0.1797 \text{ mm}$$

$$200 \text{ dw} = 1.002 \text{ mm}$$

$$\Rightarrow 0.1797 \text{ mm} = \frac{0.1797}{1.002} \times 200 \approx 36$$

5. Ans (1000.00)



$$\text{Heat} = \text{Work}$$

$$= nRT_2 \ln 2 - \frac{5}{2} nR(T_2 - T_1)$$

$$+ \frac{3}{2} nR(T_2 - T_1)$$

$$= \frac{25}{3} \times [600 \ln 2 - 300] = 1000 \text{ J}$$

6. Ans (14.08)

$$4 \times 7 - 2 \times 1 - 3 \times 2.80 = 17.6$$

$$17.6 = \frac{1}{2} \times \mu U_{\text{rel}}^2$$

$$\frac{1}{2} \times \frac{4m \times m}{4m + m} U_{\text{rel}}^2$$

$$P_1 c = \mu U_{\text{rel}}$$

$$\Rightarrow k_1 = \frac{\mu^2 U_{\text{rel}}^2}{2m} = \frac{35.2}{2} \times \frac{\mu}{m}$$

$$= 17.6 \times \frac{4}{5}$$

$$= 14.08$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (D)

Simple salt is NaCl.

2. Ans (D)

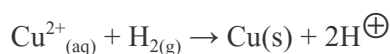
(A) Copper is extracted by Hydrometallurgy

from low grade ore.

(B) It is leached out using acid or bacteria.

* Solution is treated with scraped iron or H_2 to

recover copper from Cu^{2+} .



* Reason for using iron instead of zinc, zinc

being above iron in the electrochemical series

(Zn is more reactive), the reduction will be

faster in case zinc scraps are used. But zinc is

costlier metal than iron so using iron scraps

will be advisable and advantageous.

So (D) option is incorrect.

3. Ans (C)

Equation of process

$$\frac{\log P - 2.38}{\log V - 1.10} = \frac{2.10 - 2.38}{1.30 - 1.10} = \frac{-0.28}{0.20} = \frac{-7}{5}$$

$$\Rightarrow (\log P - 2.38) \times 5 = -7 (\log V - 1.1)$$

$$\Rightarrow 5 \log P + 7 \log V = 19.60$$

$$\Rightarrow \log P^5 + \log V^7 = 19.60$$

$$P^5 V^7 = \text{antilog}(19.60) = \text{constant}$$

$$\text{or } PV^{1.4} = \text{constant}$$

$$\therefore \gamma = 1.4 - \text{diatomic \& adiabatic.}$$

4. Ans (A)

$$pH = C\alpha = 10^{-4}$$

$$\alpha = \frac{10^{-4}}{0.0015} = \frac{1}{15}$$

$$K = G \cdot \left(\frac{\ell}{A} \right)$$

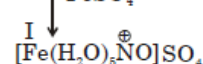
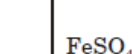
$$= 5 \times 10^{-7} \times \frac{120}{1} = 6 \times 10^{-5} \text{ Scm}^{-1}$$

$$\Lambda_m = \frac{K \times 1000}{C} = \frac{6 \times 10^{-5} \times 1000}{0.0015}$$

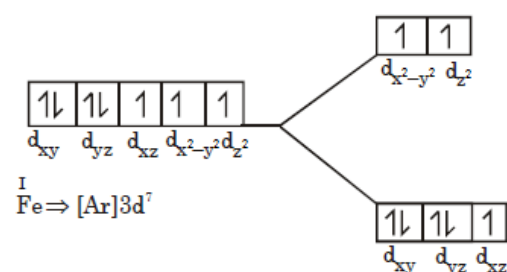
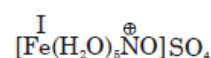
$$\Lambda_m^0 = \frac{\Lambda_m}{\alpha} = \frac{6 \times 10^{-5} \times 1000}{0.0015} \times \frac{0.0015}{10^{-4}} = 600 \text{ S cm}^2 \text{ mol}^{-1}.$$

SECTION-I (ii)

7. Ans (A,B,C,D)



Brown coloured complex



$\mu = \sqrt{15}$ B. M. brown coloured complex due to charge transfer, hybridisation $\Rightarrow sp^3d^2$

10. **Ans (A,D)**

Ans : $\rightarrow$ for ideal solution :-

$$A \rightarrow P_A^0 = 25 \text{ torr}$$

$$B \rightarrow P_B^0 = 50 \text{ Torr} \rightarrow \text{more vapour pressure}$$

$\downarrow$

means B forms more vapours and will be more in vapour phase.

so $Y_B > Y_A \rightarrow$ in vapour phase this means,

$X_B < X_A \rightarrow$ in liquid phase.

$$\text{hence } \frac{Y_B}{X_B} > \frac{Y_A}{X_A} \dots\dots\dots [A]$$

we cannot comment about the relation between

$$\frac{Y_B}{X_A} \text{ and } \frac{Y_A}{X_B} - [D]$$

[A,D]

SECTION-II

3. **Ans (68.14 to 68.15)**

$$\alpha = \frac{M_0 - M}{(n-1) \cdot M} = \frac{208.5 - 124}{(2-1) \cdot 124} = 0.681$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

Given that $\vec{p} \times \vec{q} = \vec{r}$

$$\Rightarrow (\vec{p} \times \vec{q}) \cdot \vec{r} = |\vec{r}|^2$$

$$\Rightarrow [\vec{p} \vec{q} \vec{r}] = 1$$

$$\text{Let } \vec{a} = x\vec{p} + y\vec{q} + z\vec{r} \dots\dots(1)$$

Take dot product in equation (1) both sides with $\vec{p} \times \vec{q}$, $\vec{q} \times \vec{r}$ and $\vec{r} \times \vec{p}$ we get

$$x = 1, y = 2, z = 3.$$

2. **Ans (B)**

$$\int_0^2 (4 - (4 - x^2))dx + \int_2^{2\sqrt{2}} (4 - (x^2 - 4))dx$$

$$\left(\frac{x^3}{3}\right)_0^2 + \left(8x - \frac{x^3}{3}\right)_2^{2\sqrt{2}}$$

$$\frac{8}{3} + \left(16\sqrt{2} - \frac{16\sqrt{2}}{3}\right) - \left(16 - \frac{8}{3}\right)$$

$$\frac{32\sqrt{2}}{3} - \frac{32}{3}$$

$$\frac{64\sqrt{2}}{3} - \frac{64}{3}$$

3. **Ans (C)**

$$f''(0) = 2; \lim_{x \rightarrow 0} \frac{f(5x) - 3f(3x) + 2f(2x)}{x^2}$$

Applying L.hospital twice

$$\lim_{x \rightarrow 0} \frac{25f''(5x) - 27f''(3x) + 8f''(2x)}{2} = \frac{6f''(0)}{2} = 6$$

4. **Ans (D)**

$$I = \int \frac{\operatorname{cosec}^2 x}{(\operatorname{cosec} x + \cot x)^{9/2}} dx$$

Put $\operatorname{cosec} x + \cot x = z$

$$\operatorname{cosec} x - \cot x = \frac{1}{z}.$$

$$-2\operatorname{cosec}^2 x dx = \left(1 + \frac{1}{z^2}\right) dz$$

$$\therefore I = -\frac{1}{2} \int \frac{1 + \frac{1}{z^2}}{z^{9/2}} dz = -\frac{1}{2} \left[\int z^{-9/2} dz + \int z^{-11/2} dz \right]$$

$$= -\frac{1}{2} \left[\frac{z^{-7/2}}{(-7)} + \frac{z^{-11/2}}{(-11)} \right] + C$$

$$= z^{-7/2} \left[\frac{1}{7} + \frac{z^{-3}}{11} \right] + C$$

$$= (\operatorname{cosec} x - \cot x)^{7/2} \left(\frac{1}{7} + \frac{(\operatorname{cosec} x - \cot x)^2}{11} \right) + C$$

5. **Ans (A)**

Consider $(M^2 + 2M + 2I)(M^2 - 2M + 2I)$

$$\therefore (M^2 + 2M + 2I)(M^2 - 2M + 2I) = 4I$$

$\therefore$ Neither of determinant value will be zero.

6. **Ans (A)**

$$I = \int \frac{\sqrt{x}}{(1+x^2)^{7/4}} dx = \int \frac{x^{-3}}{(1+x^{-2})^{7/4}} dx$$

Put $(1+x^{-2}) = t \Rightarrow dt = -2x^{-3} dx$

$$I = \frac{2}{3} \frac{x^{3/2}}{(x^2+1)^{3/4}} + c$$

SECTION-I (ii)

7. Ans (A,B)

$$x dy = y dx + \sin\left(\frac{1}{x}\right) dx$$

$$\frac{dy}{dx} = \frac{y}{x} + \frac{1}{x} \sin\left(\frac{1}{x}\right)$$

$$P = -\frac{1}{x}; \text{ I.F.} = e^{\log\left(\frac{1}{x}\right)} = \frac{1}{x}$$

$$\frac{y}{x} = \int \frac{1}{x^2} \sin\left(\frac{1}{x}\right) dx$$

$$= - \int \sin t dt = \cos \frac{1}{x} + c$$

$$y = x \left(\cos \frac{1}{x} + c \right)$$

$$y = x \left(\cos \frac{1}{x} + 3 \right) \text{ is on such}$$

$$y = x \left(\underbrace{\cos \frac{1}{x} + 1}_{2} + 2 \right)$$

$$= x \left(2 \cos^2 \frac{1}{2x} + 2 \right)$$

$$= 2x \left(2 \cos^2 \frac{1}{2x} + \sin^2 \frac{1}{2x} \right)$$

8. Ans (A,C,D)

$$f(0) = 0, f(1) = 2, f(2) = 4$$

Using L.M.V.T on [0,1]

$\exists$ a point $x = \alpha \in (0,1)$, such that

$$f'(\alpha) = \frac{f(1) - f(0)}{1 - 0} = 2$$

Similarly apply L.M.V.T. on $y = f(x)$ in [1,2]

$\exists$ a point $x = \beta \in (1,2)$, such that

$$f'(\beta) = \frac{f(2) - f(1)}{2 - 1} = 2$$

Now, Applying Rolle's theorem on $y = f'(x)$

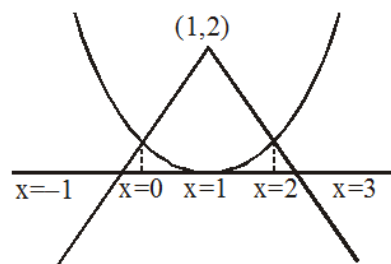
$$x \in [\alpha, \beta]$$

$\therefore f'(\alpha) = f'(\beta)$ and other conditions are satisfied

$\exists$ a point $x = \gamma \in (\alpha, \beta)$ $f''(\gamma) = 0$ for some

$$\gamma \in (0,2)$$

9. Ans (B,C)



Area bounded by f & g

$$= 2 \int_0^1 ((x+1) - (x-1))^2 dx$$

$$= 2 \left(\frac{(x+1)^2}{2} - \frac{(x-1)^3}{3} \right) \Big|_0^1 = \frac{7}{3}$$

Area bounded by $f(x)$, $g(x)$ and x -axis

$$= 4 - \frac{7}{3} = \frac{5}{3}$$

10. Ans (A,B,C,D)

$$|\vec{v}_1 + \vec{v}_2 - \vec{v}_3|^2 + |\vec{v}_2 + \vec{v}_3 - \vec{v}_1|^2 + |\vec{v}_3 + \vec{v}_1 - \vec{v}_2|^2 = 36$$

$$\Rightarrow \vec{v}_1 + \vec{v}_2 + \vec{v}_3 = \vec{0}$$

So, $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are coplanar vectors.

11. Ans (A,B,C,D)

Given that $AA^T = I$ and $BB^T = I$

(A) Let $C = A^2$

$$CC^T = AAA^T A^T = I$$

(B) Let $C = AB$

$$CC^T = ABB^T A^T = I$$

(C) Let $C = AABBB$

$$CC^T = AABBB^T B^T A^T A^T = I$$

(D) Let $C = A^T B$

$$CC^T = A^T B B^T A = I$$

12. Ans (A,C)

$$P_n = \lim_{x \rightarrow 0^+} \log \left(\frac{\tan x}{x} \cdot \frac{\tan 2x}{x} \cdot \frac{\tan 3x}{x} \cdots \frac{\tan nx}{x} \right)$$

$$= \log n!$$

$$P_{256} - P_{255} = \log 256 = 8 \log 2!$$

$$\therefore k = 8 \text{ and } n = 2$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (15.00)**

Clearly $a_n, b_n \geq 0 \forall n \in \mathbb{N}$

$$\because 1 - x \leq 1 - x^2 \leq 1 \forall x \in [0, 1]$$

$$\Rightarrow \int_0^1 (1-x)^n dx \leq \int_0^1 (1-x^2)^n dx \leq \int_0^1 dx$$

$$\Rightarrow \frac{1}{n+1} \leq a_n \leq 1$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} \right)^{\frac{1}{n}} \leq \lim_{n \rightarrow \infty} (a_n)^{\frac{1}{n}} \leq 1$$

$$\text{but } \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} \right)^{\frac{1}{n}} = e^{\lim_{n \rightarrow \infty} \frac{-\ln(n+1)}{n}} = e^0$$

(by L' Hopital)

$$\because \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = 1 \text{ similarly } \lim_{n \rightarrow \infty} \sqrt[n]{b_n} = 1$$

$$\text{Therefore, } \lim_{n \rightarrow \infty} (10\sqrt[n]{a_n} + 5\sqrt[n]{b_n}) = 15$$

2. **Ans (1.00)**

Order = 1

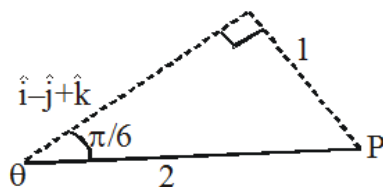
$$\Rightarrow \left(\frac{dy}{dx} \right)^2 + \frac{dy}{dx} + 3 \frac{dy}{dx} (x) = x^3$$

Degree = 2

3. **Ans (3.14)**

$$(\hat{i} - \hat{j} + \hat{k}) \cdot (a\hat{i} + b\hat{j} + c\hat{k}) = 3$$

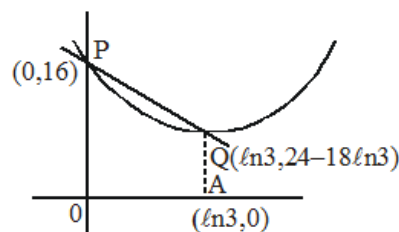
$$\cos \theta = \frac{\sqrt{3}}{2}$$



$$\sqrt{3} \cdot 2 \cos \theta = 3 \{ \because a^2 + b^2 + c^2 = 4 \}$$

locus of 'P' is circle with radius $A = \pi$

4. **Ans (1.50)**



$$\text{Area of trap OAQP} = \frac{1}{2} (40 - 18\ln 3) \cdot 4\ln 3$$

$$\begin{aligned} \text{Req. area} &= \frac{1}{2} (40 - 18\ln 3) \cdot 4\ln 3 - \int_0^{4\ln 3} (e^{2x} - 18x + 15) dx \\ &= 5\ln 3 - 4 \end{aligned}$$

5. **Ans (0.50)**

$$2A + B = B^T - A^T \quad \dots(i)$$

take transpose both side

$$2A^T + B^T = B - A \quad \dots(ii)$$

equation (i) - equation (ii)

$$\Rightarrow 3A = -3A^T \Rightarrow A^T = -A$$

$$\Rightarrow 2A + B = B^T + A$$

$$\Rightarrow A + B = B^T$$

$$\Rightarrow |A + B| = |B^T| = |B| = 1$$

6. **Ans (9.00)**

$$\text{Let } \left(\frac{A}{3} \right) = B$$

$$|\text{adj} B^{-1}| = |B^{-1}|^2 = \frac{1}{|B|^2} = \frac{1}{\left| \frac{A}{3} \right|^2}$$

$$= \frac{3^6}{|A|^2} = \frac{3^6}{9^2} = 3^2 = 9$$

JEE(Main + Advanced) : Enthusiast Course (PHASE : TEAS)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C	A,C,D	A,B,D	A,C	A,B,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	15.62 to 15.63	1.75	640.00	75.00	2.65 to 2.66	2.70 to 2.71
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	5	1	9	7	7

PART-2 : CHEMISTRY

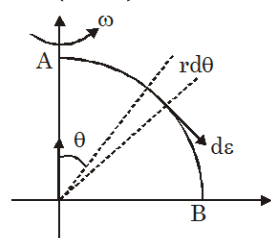
SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	A,B,D	A,B,D	B,C,D	A,D	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	281.00	18.00	231.00	5.00	5.00	3.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	2	3	1	3	5

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	C	A,C,D	A,C	A,B,D	A,B	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.73	3.46	1.20	0.25	2.00	3.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	1	4	7	2	6

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (A,C)



$$d\varepsilon = B \perp v \ell \perp$$

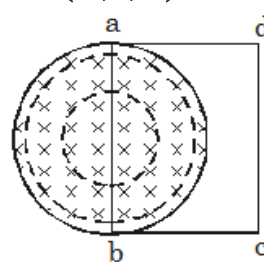
$$= B_r r d\theta \omega \sin \theta$$

$$\varepsilon = \frac{2\mu_0 m \omega r^2}{4\pi r^3} \int_0^{\pi/2} \sin \theta \cos \theta d\theta$$

$$= \frac{2 \times 10^{-7} \times 0.4 \times 400}{1 \times 2} \times -\frac{\cos 2\theta}{2} \Big|_0^{\pi/2}$$

$$= 16 \times 10^{-6} \times [1] = 16\mu\text{V}$$

2. Ans (A,C,D)



$$\varepsilon_{ab} = - \int \vec{E} \cdot d\vec{\ell} = 0$$

$$\varepsilon_{adcb} = \frac{\pi r^2}{2} \times \frac{dB}{dt}$$

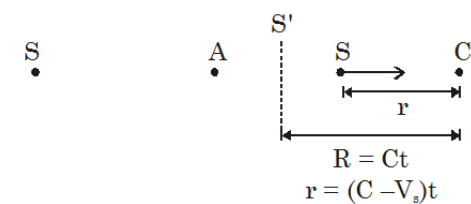
$$= 0.1 \times \pi \times \frac{0.01}{2}$$

$$= \frac{\pi}{2} \text{ mV}$$

$$i = \frac{\pi}{2 \times 40} \text{ mA}$$

$$V_{AB} = iR = i \times 10\Omega = \frac{\pi}{8} \text{ mV}$$

3. Ans (A,B,D)



60m/s

$$f_A = f_0$$

$$f_B = \frac{340 + 60}{340} f_0$$

$$f_C = \frac{340}{340 - 60} f_0$$

$$\Rightarrow f_C > f_B > f_A$$

$$I_A = I_0 = \frac{E_{1 \text{ packet}}}{T' \times 4\pi r^2}$$

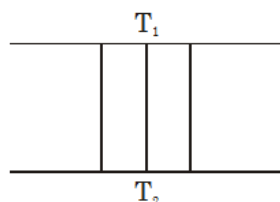
$$I_B = \frac{E_{1 \text{ packet}}}{T' \times 4\pi r^2} = I_A \times \frac{f_B}{f_0} = I_A \times \frac{400}{340}$$

For C, wave was emitted from a larger distance

$$I_C = \frac{E_{1 \text{ packet}}}{T'' \times 4\pi R^2} = I_A \times \frac{f_C}{f_0} \times \left(1 - \frac{V_s}{C}\right)^2$$

$$= I_A \left(1 - \frac{V_s}{C}\right)$$

4. Ans (A,C)

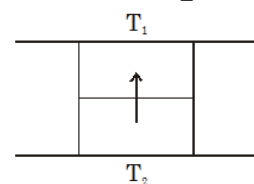


$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_1 = \frac{\ell}{K_1 A} \quad R_2 = \frac{\ell}{K_2 A}$$

$$R = \frac{\ell}{K_{eq} \times 2A}$$

$$\Rightarrow K_{eq} = \frac{K_1 + K_2}{2}$$



$$R = R_1 + R_2$$

$$\frac{2\ell}{K_{eq} A} = \frac{\ell}{K_1 A} + \frac{\ell}{K_2 A}$$

$$\Rightarrow K_{eq} = \frac{2K_1 K_2}{K_1 + K_2}$$

5. Ans (A,B,C)

$$\beta = \frac{\lambda D}{d}$$

$$\Rightarrow \lambda = \frac{\beta d}{D} = \frac{0.25 \times 10^{-3} \times 2 \times 10^{-3}}{1} = 500 \text{ nm}$$

$$\frac{\Delta \lambda}{\lambda} = \frac{\Delta \beta}{\beta} + \frac{\Delta d}{d} + \frac{\Delta D}{D}$$

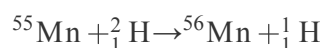
$$= \frac{0.01}{0.25} + \frac{0.01}{2.00} + \frac{0.01}{1.00}$$

$$\frac{\Delta \lambda}{\lambda} = 4 + 0.5 + 1 = 5.5\%$$

near central maxima

$$d \sin \theta \approx d \tan \theta$$

6. Ans (A,B,C)



$$\text{Steady state} \Rightarrow \frac{dN}{dt} = \frac{dN}{dt}$$

$$\Rightarrow \frac{dN}{dt} = \lambda N_0 = 6.93 \times 10^{10}$$

$$= \frac{0.693}{3 \times 3600} N_0 = 6.93 \times 10^{10}$$

$$N_0 = 10800 \times 10^{11}$$

$$m = \frac{N_0}{N_a} \times 56 \text{ gm}$$

$$= \frac{10800 \times 10^{11} \times 56}{6 \times 10^{23}} \approx 100 \text{ ng}$$

SECTION-II

1. Ans (15.62 to 15.63)

$$\varepsilon = A \frac{dB}{dt} = \pi r^2 \times \pi$$

$$i = \frac{\varepsilon}{\frac{\rho \ell}{A}} = \frac{\pi^2 r^2}{\rho \times 2\pi r} \times A$$

$$2\pi r A = V = \frac{m}{d} = \frac{1}{8} \times 10^{-3}$$

$$\Rightarrow i = \frac{1}{16} \times \frac{10^{-3}}{2 \times 2} = \frac{1000}{64} \mu\text{A}$$

$$= 15.625 \mu\text{A}$$

2. Ans (1.75)

$$f = \frac{1}{2 \times 0.9} \sqrt{\frac{mg}{\mu}} = \frac{1}{1.8} \sqrt{\frac{9Vg}{\mu}}$$

$$f = \frac{1}{2 \times 0.7} \sqrt{\frac{mg - \rho_A V g}{\mu}} = \frac{1}{1.4} \sqrt{\frac{(9 - \rho_A) V g}{\mu}}$$

$$f = \frac{1}{2 \times 0.5} \sqrt{\frac{mg - \rho_B V g}{\mu}} = \frac{1}{1} \sqrt{\frac{(9 - \rho_B) V g}{\mu}}$$

$$\Rightarrow 1.8 = \sqrt{\frac{9}{9 - \rho_B}}$$

$$\frac{1.8}{1.4} = \sqrt{\frac{9}{9 - \rho_A}}$$

$$\Rightarrow 1 - \frac{\rho_A}{9} = \frac{196}{324} \Rightarrow \rho_A = 9 \times \frac{128}{324}$$

$$1 - \frac{\rho_B}{9} = \frac{100}{324} \Rightarrow \rho_B = 9 \times \frac{224}{324}$$

$$\frac{\rho_B}{\rho_A} = \frac{224}{128} = 1.75$$

3. Ans (640.00)

$$h = 25 \text{ cm}$$

$$f = \frac{u}{2\ell} = 640 \text{ Hz}$$

4. Ans (75.00)

$$V = K_1 \left(\frac{PV}{nR} \right)^3$$

$$\Rightarrow P^3 V^2 = K$$

$$\Rightarrow P V^{2/3} = K$$

$$C = \frac{3}{2}R - \frac{R}{\frac{2}{3} - 1} = 4.5R$$

$$H = nC\Delta T$$

$$= 9 \times \frac{25}{3} = 75 \text{ J}$$

5. Ans (2.65 to 2.66)

$$\ell = 10.1 \text{ mm} = 1.01 \text{ cm}$$

$$d = \frac{2.736}{(1.01)^3} = 2.655$$

6. Ans (2.70 to 2.71)

$$\text{Extra energy} = 3.4 + 2 = 5.4 \text{ eV}$$

$$K_{\max} = 5.4 - \frac{1240}{460} = 2.70 \text{ to } 2.71 \text{ eV}$$

SECTION-III

1. Ans (5)

$$T = 2\pi \sqrt{\frac{\ell}{gh}} = 2\pi \sqrt{\frac{\ell}{g \left(1 - \frac{2h}{R} \right)}}$$

$$= 2\pi \sqrt{\frac{\ell (1 + \alpha \times 10)}{g}}$$

$$= 1 + 10\alpha = \frac{1}{\left(1 - \frac{2h}{R} \right)} \approx 1 + \frac{2h}{R}$$

$$\Rightarrow \alpha = \frac{h}{5R}$$

2. Ans (5)

$$8 \times 0.5 \times (0 - (-20)) + mL = 8 \times 1 \times (60 - 0)$$

$$m = \frac{480 - 80}{80} = 5 \text{ kg}$$

3. Ans (1)

$$\sqrt{2.8 - 2.7T} = \sqrt{0.09} = 0.3$$

4. Ans (9)

$$I_{\max} = (\sqrt{I_1} + \sqrt{0.64I_1}) = 3.24I_1$$

$$I_{\min} = (\sqrt{I_1} - \sqrt{0.64I_1})^2 = 0.04I_1$$

$$\sqrt{\frac{I_{\max}}{I_{\min}}} = 9$$

5. Ans (7)

$$\frac{\lambda_e D}{d} = 10^{-7} \Rightarrow \lambda_e = \frac{2.21 \times 10^{-10}}{2.6}$$

$$\frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} v} = \frac{2.21 \times 10^{-10}}{2.6}$$

$$v = \frac{6}{7} \times 10^7 \text{ m/s}$$

$$E = \frac{hc}{\lambda} = \frac{1}{2}mv^2$$

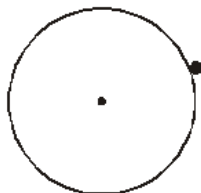
$$= \frac{1}{2} \times 9.1 \times 10^{-31} \times \frac{36}{49} \times 10^{14}$$

$$\lambda = 6.63 \times 10^{-34} \times 3 \times 10^8 \times \frac{98}{36 \times 9.1} \times 10^{17}$$

$$\beta = \frac{\lambda D}{d} = \frac{6.63 \times 98}{12 \times 9.1} \times 10^{-9} \times \frac{2.6}{2.21 \times 10^{-3}}$$

$$= 7 \mu\text{m}$$

6. Ans (7)



$$r = R_0 A^{1/3} = 1.2 \times 10^{-15} \times 4 = 4.8 \times 10^{-15}$$

$$F = \frac{9 \times 10^9 \times 28 \times (1.6 \times 10^{-19}) \times 1.6 \times 10^{-19}}{4.8 \times 10^{-15} \times 4.8 \times 10^{-15}}$$

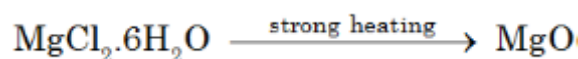
$$= 280 \text{ N}$$

$$\frac{F}{40} = 7$$

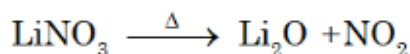
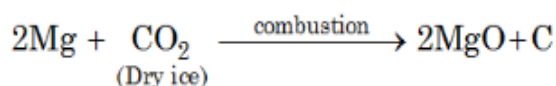
PART-2 : CHEMISTRY

SECTION-I

1. Ans (A,B,D)



(MgO reacts with water crystallisation)



SECTION-II

1. Ans (281.00)

Net stabilization energy of Fe^{2+} with CN^-

$$\text{ligands } t_{2g}^{2,2,2} e_g^{0,0} = -2.4 \Delta_0 + 2P$$

$$= (-2.4 \times 25000 + 2 \times 15000) \text{ cm}^{-1}$$

$$= -30000 \text{ cm}^{-1}$$

net stabilisation energy of Fe^{2+} ions with

$$\text{H}_2\text{O ligands } t_{2g}^{2,1,1} e_g^{1,1} = -0.4 \Delta_0$$

$$= 4000 \text{ cm}^{-1}$$

so required difference = 26000 cm^{-1}

$$= 26 \times 10^5 \text{ m}^{-1}$$

$$= 6 \times 10^{-34} \times 3 \times 10^8 \times 26 \times 10^5 \times 6 \times 10^{23}$$

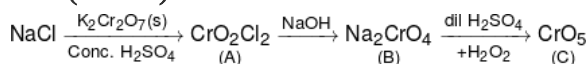
$$\times 10^{-3} \text{ kJ/mole}$$

$$= 2808 \times 10^{-1} \text{ kJ/mole}$$

$$= 281 \text{ kJ/mole}$$

$$= 280.8 \text{ kJ/mol}$$

2. Ans (18.00)



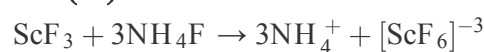
total number of atoms in A, B & C = $5 + 7 + 6 = 18$.

4. Ans (5.00)

5

SECTION-III

2. Ans (2)



hybridisation d^2sp^3 ($d_{x^2-y^2}$ & d_{z^2})

3. Ans (3)

$$G = H - TS$$

As process is reversible adiabatic the entropy stays constant.

$$\Rightarrow dG = dH - SdT$$

$$\Delta G = \Delta H - S\Delta T = nC_{p,m}\Delta T - S\Delta T = 1 \times \frac{7}{2} \times R\Delta T - S\Delta T$$

$$= \frac{7}{2} \times 2 \times \Delta T - 3\Delta T = 4 \times \Delta T$$

for adiabatic process

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T_2 = T_1 \times \left(\frac{V_1}{V_2}\right)^{\gamma-1} = 1000 \times \left(\frac{1}{32}\right)^{7/5-1} = 250 K$$

$$\Delta T = 250 - 1000 = -750$$

$$\Rightarrow \Delta G = 4 \times 4T = 4 \times -750 = -3000 \text{ cal}$$

$$\Rightarrow |\Delta G| = 3 \text{ kcal}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$\therefore (1 + \sqrt{2})^n = A_n + B_n \sqrt{2}$$

$$\Rightarrow (1 - \sqrt{2})^n = A_n - B_n \sqrt{2}$$

$$\text{on solving, } A_n = \frac{(1 + \sqrt{2})^n + (1 - \sqrt{2})^n}{2},$$

$$B_n = \frac{(1 + \sqrt{2})^n - (1 - \sqrt{2})^n}{2\sqrt{2}}$$

$\therefore$

$$\lim_{n \rightarrow \infty} \frac{A_n}{B_n} = \lim_{n \rightarrow \infty} \sqrt{2} \cdot \left(\frac{1 + \left(\frac{1-\sqrt{2}}{1+\sqrt{2}}\right)^n}{1 - \left(\frac{1-\sqrt{2}}{1+\sqrt{2}}\right)^n} \right) = \sqrt{2}$$

$$\therefore \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^n \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\& (A) \rightarrow \text{limit} = 1 \quad (C) \rightarrow \text{limit} = \sqrt{2}$$

$$(B) \rightarrow \text{limit} = 0$$

2. Ans (A, C, D)

$$A^T = A \& B^T = -B$$

$$(A) (ABA^T)^T = AB^T A^T$$

$$= -(ABAT)$$

$$(B) AB^T + BA^T = -AB + BA$$

$$(BA - AB)^T = (BA)^T - (AB)^T$$

$$= A^T B^T - B^T A^T$$

$$= -AB + BA$$

$$(C) ((A + B)(A - B))^T = (A - B)^T (A + B)^T$$

$$= (A^T - B^T)(A^T + B^T)$$

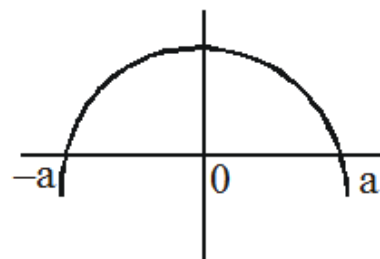
$$= (A + B)(A - B)$$

$$(D) ((A + I)(B - I))^T = (B^T - I)(A^T + I)$$

$$= -(B + I)(A + I)$$

3. Ans (A, C)

$$ay = 3(a - x)(a + x)$$



$$\text{Area} = 2 \int_0^a \frac{3}{a} (a^2 - x^2) dx = 64$$

$$\Rightarrow \frac{3}{a} \left(\left(a^2 x - \frac{x^3}{3} \right) \right)_0^a = 32$$

$$\Rightarrow \frac{3}{a} \left(a^3 - \frac{a^3}{3} \right) = 32$$

$$\Rightarrow 3 \left(a^2 - \frac{a^2}{3} \right) = 32 \Rightarrow 3a^2 - a^2 = 32$$

$$\Rightarrow 2a^2 = 32 \Rightarrow a^2 = 16 \Rightarrow a = \pm 4$$

4. Ans (A,B,D)

$$B = \text{adj}A$$

$$\therefore B^T B^{-1} = A \Rightarrow |A| = |B^T| |B^{-1}| = 1$$

take adjoint on both sides

$$\Rightarrow \text{adj}B^{-1} \cdot \text{adj}B^T = \text{adj}A$$

$$\Rightarrow (\text{adj}B)^{-1}(\text{adj}B)^T = B$$

$$\Rightarrow (\text{adj}B)^T = (\text{adj}B)B$$

$$\Rightarrow (\text{adj}B)^T = |B|I$$

$$\Rightarrow \text{adj}B = |B|I$$

$$\Rightarrow \text{adj}(\text{adj}A) = |\text{adj}A|I$$

$$\Rightarrow |A|^{n-2}A = |A|^{n-1}I$$

$$\Rightarrow A = |A|I$$

$$\Rightarrow A = I (\because |A| = 1)$$

$$\Rightarrow B = I$$

5. Ans (A,B)

$$V = |\vec{a} \cdot (\vec{b} \times \vec{c})| = |\vec{a}| |\vec{b} \times \vec{c}| |\cos \alpha|$$

$$\text{Similarly, } V = |\vec{b}| |\vec{c} \times \vec{a}| |\cos \alpha| \quad \text{and}$$

$$V = |\vec{c}| |\vec{a} \times \vec{b}| |\cos \alpha|$$

$$\text{While } S = 2 \left(|\vec{a} \times \vec{b}| + |\vec{b} \times \vec{c}| + |\vec{c} \times \vec{a}| \right)$$

$$= 2 \left(\frac{V}{|\vec{c}| |\cos \alpha|} + \frac{V}{|\vec{a}| |\cos \alpha|} + \frac{V}{|\vec{b}| |\cos \alpha|} \right)$$

$$\Rightarrow \frac{V}{S} = \frac{|\cos \alpha|}{2} \left[\frac{1}{|\vec{a}|} + \frac{1}{|\vec{b}|} + \frac{1}{|\vec{c}|} \right]^{-1}$$

$$\text{Now, } V = |\vec{a} \cdot (\vec{b} \times \vec{c})| = |\vec{a}| |\vec{b} \times \vec{c}| |\cos \alpha|$$

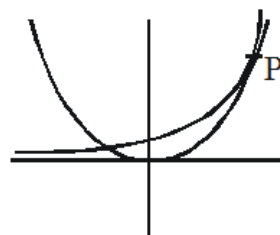
$$= |\vec{a}| |\vec{b}| |\vec{c}| \sin(\hat{b} \wedge \hat{c}) |\cos \alpha| \leq |\vec{a}| |\vec{b}| |\vec{c}|$$

$$\text{Also, } \sin(\vec{a} \wedge \vec{b}) = \sin(\vec{b} \wedge \vec{c}) = \sin(\vec{c} \wedge \vec{a})$$

So, it not necessary that they are equally inclined.

6. Ans (B,C,D)

Since both the curves touches each other



$$\Rightarrow m_1 = m_2$$

$$4(x-a) = 2e^{2x}$$

$$4(x-a) = 2.2(x-a)^2$$

$$\Rightarrow (x-a)(x-a-1) = 0$$

$$\Rightarrow x = a \text{ or } x = a+1$$

$$\text{If } x = a \text{ then } (I^{\text{st}}) \text{ curve} \Rightarrow y = 0$$

$$\& 2^{\text{nd}} \text{ curve} \Rightarrow y = e^{2a} \text{ but } e^{2a} \neq 0 \forall a \in \mathbb{R}$$

$$\therefore x \neq a$$

$$\Rightarrow x = a+1$$

$$\therefore 1^{\text{st}} \text{ curve} \Rightarrow y = 2 \quad \therefore (x,y) = (a+1, 2)$$

$$(x,y) \text{ lies on } 2^{\text{nd}} \text{ curve also, } \therefore 2 = e^{2a+2}$$

$$\Rightarrow 2a+2 = \ln 2$$

$$a = \frac{1}{2}(\ln 2 - 2)$$

$$\Rightarrow -1 < a < 0$$

SECTION-II

1. Ans (1.73)

$$V_{\text{DABC}} = \frac{1}{6} = \frac{1}{3} \times (\text{area of base}) \times \text{height}$$

Also,

$$\frac{1}{6} \leq \frac{1}{3} \times \frac{1}{2} \times AC \times BC \times \sin 45^\circ \times AD$$

$$\therefore \left(\frac{AC}{\sqrt{2}} \right) \cdot BC \cdot AD \geq 1$$

$$\text{Now, } \frac{AC}{\sqrt{2}} + BC + AD \geq 3 \cdot \sqrt[3]{\frac{AC}{\sqrt{2}} \cdot BC \cdot AD}$$

(using A.M.- G.M.)

$$\therefore \frac{AC}{\sqrt{2}} \cdot BC \cdot AD \leq 1$$

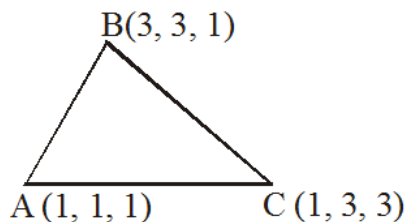
$\therefore$ Equality must hold

$$\frac{AC}{\sqrt{2}} = BC = AD = 1$$

$\therefore$ AD must be perpendicular

$$\therefore CD^2 = AD^2 + AC^2 = 3$$

2. Ans (3.46)



And ΔABC is an equilateral triangle with side
 $= 2\sqrt{2}$

$$\therefore \text{Area} = \frac{\sqrt{3}}{4} \times 8 = 2\sqrt{3}$$

3. Ans (1.20)

$$A^2 = I \Rightarrow A^4 = A^6 = I$$

$$2A^2 + 4A^4 + 6A^6 = 12I_3$$

4. Ans (0.25)

$$I = \int_0^{\pi/4} \left(\frac{\sin^2 7\theta - \sin^2 5\theta}{\sin 2\theta} + \frac{\sin^2 5\theta - \sin^2 3\theta}{\sin 2\theta} + \frac{\sin^2 3\theta - \sin^2 \theta}{\sin 2\theta} + \frac{\sin^2 \theta}{\sin 2\theta} \right) d\theta$$

$$= \int_0^{\pi/4} \left[\sin 12\theta + \sin 8\theta + \sin 4\theta + \frac{1}{2} \tan \theta \right] d\theta$$

$$= - \left[\frac{\cos 12\theta}{12} + \frac{\cos 8\theta}{8} + \frac{\cos 4\theta}{4} \right]_0^{\pi/4} + \frac{1}{2} [\log |\sec \theta|]_0^{\pi/4}$$

$$= - \left[\frac{-1-1}{12} + \frac{1-1}{8} + \frac{-1-1}{4} \right] + \frac{1}{2} (\log \sqrt{2})$$

$$= \frac{1}{6} + \frac{1}{2} + \frac{1}{4} \log 2 = \frac{2}{3} + \frac{1}{4} \log 2$$

5. Ans (2.00)

$$\alpha y^2 = 2x + \alpha$$

$$\alpha = \frac{2x}{y^2 - 1}, \text{ differentiate}$$

$$0 = (y^2 - 1) 2 - 2x \cdot 2y \frac{dy}{dx}$$

$$2xy \frac{dy}{dx} - y^2 + 1 = 0$$

$$d = 1, \quad O = 1$$

$$d + O = 2$$

6. Ans (3.00)

$$\Rightarrow I_1 = \int_0^{\pi} (\pi - x) \sin x e^{\cos 4x} dx$$

(King)

$$2I_1 = \pi \int_0^{\pi} \sin x \cdot e^{\cos 4x} dx$$

$$2I_1 = \pi \int_0^{\pi/2} [\sin x \cdot e^{\cos 4x} + \sin x e^{\cos 4x}] dx$$

(Queen)

$$I_1 = \pi \int_0^{\pi/2} \sin x \cdot e^{\cos 4x} dx$$

$$= \pi \int_0^{\pi/2} \cos x \cdot e^{\cos 4x} dx$$

(King)

$$\left[\frac{I_1}{I_2} \right] = 3$$

SECTION-III

1. Ans (1)

Given,

$$2xy \frac{dy}{dx} = x^2 + y^2$$

$$\Rightarrow \frac{2xydy - y^2dx}{x^2} = dx$$

$$\Rightarrow d \left(\frac{y^2}{x} \right) = dx$$

Integrating, $\frac{y^2}{x} = x + c$, since it passes through (2,1)

$$\Rightarrow c = -\frac{3}{2}$$

$$\therefore y^2 = x^2 - \frac{3}{2}x \Rightarrow \left(x - \frac{3}{4} \right)^2 - y^2 = \frac{9}{16}$$

Clearly hyperbola is rectangular, so its eccentricity $e = \sqrt{2}$

$$\therefore e^2 - 1 = 1$$

2. **Ans (1)**

$$\begin{aligned}
 f' \left(\frac{\pi}{3} \right) &= 0 + 0 + \left(\sin \frac{\pi}{3} \right)^{1/3} \cdot \left(\sin \frac{2\pi}{3} \right)^{1/2} \cdot \left(\cos \frac{3\pi}{3} \right) \cdot 3 \\
 &= \left(\frac{\sqrt{3}}{2} \right)^{1/3} \cdot \left(\frac{\sqrt{3}}{2} \right)^{1/2} \cdot (-3) \\
 &= \frac{-3^{17/12}}{2^{5/6}}
 \end{aligned}$$

3. **Ans (4)**

(using L' Hopitals rule)

On differentiating, we get

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{(\tan^{-1} x) \cdot x^2}{(\sec^2 x - 1) \sec x \tan x} &= \lim_{x \rightarrow 0} \frac{\tan^{-1} x \cdot x^2}{\sec x \tan^3 x} \\
 &= 1
 \end{aligned}$$

4. **Ans (7)**

$$\begin{aligned}
 L &= \lim_{n \rightarrow \infty} \left(\frac{n+2}{n^2+n+1} + \frac{n+4}{n^2+2n+2^2} + \dots + \frac{n+2(2n)}{n^2+(2n)n+(2n)^2} \right) \\
 &= \lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \frac{n+2r}{n^2+rn+r^2} \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{1+2\left(\frac{r}{n}\right)}{\left(\frac{r}{n}\right)^2 + \left(\frac{r}{n}\right) + 1} \\
 &= \int_0^2 \frac{1+2x}{x^2+x+1} dx = \ln(x^2+x+1)_0^2 = \ln 7 \\
 e^L &= 7
 \end{aligned}$$

5. **Ans (2)**

$$\begin{aligned}
 A &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \text{tr}(A) = a + d \\
 A^2 &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix} \\
 \text{tr}(A^2) &= a^2 + 2bc + d^2 \\
 -(a^2 + 2bc + d^2) + (a + d)^2 &= -\text{tr}(A^2) + (\text{tr}(A))^2 \\
 2(-bc + ad) &= 2|A|
 \end{aligned}$$

6. **Ans (6)**

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $n(A)$ denotes number of such matrices satisfying given conditions.

$$\text{Now, } \text{adj} A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix}$$

$$\Rightarrow d = -a$$

$$\text{Now, } |A| = ad - bc = -1$$

$$\Rightarrow a^2 + bc = 1$$

Case I : $a = 0$

$$\text{so, } d = 0 \text{ and } bc = 1$$

$$\Rightarrow n(A) = 2$$

Case II : $a = 1$ or -1

$$\text{so, } d = -1 \text{ or } 1, \text{ and } bc = 0$$

$$n(A) = 2 \times 5$$

$$\text{hence, } n(A) = 12$$